NAME :_____NUMBER_____

MTENDERE SECONDARY SCHOOL

2018/2019 MOCK EXAMINATIONS

MATHEMATICS

PAPER II

(100 Marks)

Thursday, 4 April **Time Allowed: 2 hrs 30 min**
08: 00 - 10: 30 AM

Instructions:

1. **This paper contains 18 printed pages. Please check.**
2. Answer **all** questions in **Section A** and three questions in **Section B**
3. Write your **Name** and **Examination number** on top of each page of your question paper.
4. All answers should be written in the spaces provided after every question
5. Calculators and four figure tables may be used
6. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
TOTAL			

NAME :_____NUMBER_____

1. a) Without using table or calculators , evaluate

$$\sqrt{\frac{\log 4096}{1+\ \frac{1}{2}\log 0.16+\ \frac{1}{3}\log 8}}$$

(5 marks)

b) Find the equation of straight line with gradient 3 passes through the point A (3, -4)

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(5 marks)

2. a) P varies directly as cube of t and inversely as the square root of n. When P = 16, t = 2 and n = 9. Find the n in terms of P and t

(4 marks)

b) Find the value of x and y if $(x \quad y)\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = (9 \quad 6)$

(5 marks)

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3. a) **Figure 1** , the function f: x $\rightarrow$ $x^2 - 2x - 1$ is defined on the domain {-1, 0, 1, 2). Copy and complete the mapping diagram for.

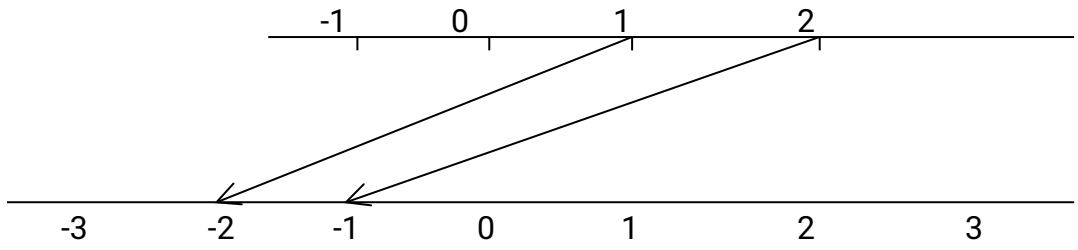

Figure 1

(4 marks)

b) Make x the subject of the formula

$$y = \frac{p}{2x^3 \ q}$$

(5 marks)

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4. a) Draw two circles of radii 3.5 cm and 2.5 cm such that their centre are 8.5 cm apart. Construct a transverse common tangent. Measure its length.

(4 marks)

b) Points **A, B**, and **C** are on the circle such that **ABC** = 108^0. Find the angle of between the tangent at **A** and **C**

(4 marks)

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5. In **Figure 2** , **ABCD** is a circle and **DE** is a tangent to the circle at **D** . **AC** is parallel to the tangent **DE**

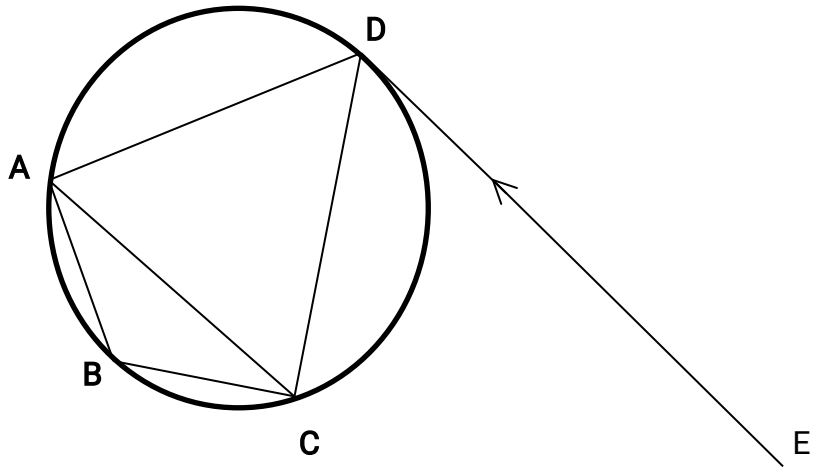

Figure 2

Prove that :

i) Triangle **ADC** is isosceles triangle (2 marks)

ii) angle **ABC** is twice angle **ADC** **(3 marks)**

HELP

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b - Express $\frac{1}{3\sqrt{2-3}}$ with a rational denomination in its simplest form.

(4 marks)

6. a) The variance of two temperature measurements in degree Celsius 2 and 2a is 9 . Calculate the possible values of a

(5 marks)

b) Make d the subject of the formula $S = \frac{n}{2}[2a + (n - 1)d]$

(5 marks)

SECTION B (45 Marks)

Answer any **three** questions

7. i) Copy and complete the table below for the relations $y = 5 - x$

x	-3	-2	0	1	2	3	5
y			5			2	0

Use your table to draw the graph of $y = 5 - x$

ii. Make a table of values for the relations $y = x^2 - 1$ for $-3 \leq x \leq 3$. Hence draw the graph of $y = x^2 - 1$.

(7 marks)

b) In the acute – angled triangle **ABC** , **AB** = 6 cm, **AC** = 4 cm and **N** is the foot of the perpendicular from **A** to the side **BC**. Show that $BN^2 - NC^2 = 2O$.

HELP

BOTH

(8 marks)

8. Town **X** is 10 km due north of town **Y**.

a) The bearing of ship **H** from town **X** is $145^0\ 34'$ **(S34 26′E)** and town **Y** is $O55^0\ 34'$ **(N550 34′E).** How far is the ship from **Y**?

(7 marks)

b) The time **(T)** it takes to enter a stadium is partly constant and partly varies as the number of people **(N)** entering the stadium. If there are four people it takes 12 seconds to enter the stadium and if there are 5 people it takes 14 seconds. How long will it take , if there are 28 people entering the stadium

(8 marks)

9. a) The areas of two similar triangle are 64 cm^2 and 36 cm^2 . The base of the bigger triangle is 16 cm. Calculate the corresponding height of the smaller triangle.

(7 marks)

b) Solve the simultaneous equations

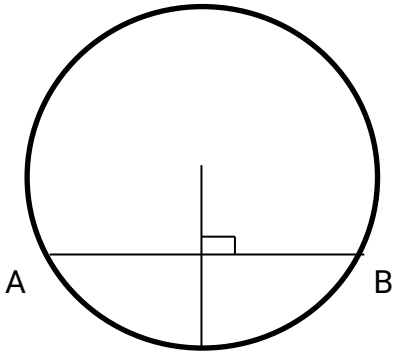
$x - 3y - 1 = 0$

$x^2 - 8y^2 - y + 5 = 0$

(8 marks)

10. **Figure 3** shows a circle **ACB** centre **O** .

a) **OM** = 3x cm , **MC** = 2x cm and angle **OMB** = 90^0



Find **AB** interms of x

(7 marks)

b) A person running at a speed of y metres per second passes a stationery train 50

metres long. If the person takes $\frac{25}{y-1}$ seconds to pass the train, calculate the value of y.

(8 marks)

11. a) Using a ruler and a pair of compass only, construct, in the same diagrams .

i. A circle centre O of radius 3 cm.

ii. Two tangents to the circle from a point **T** which is 8 cm from O touching the circle at the circle at points **M** and **N**.

iii. Measure and state the angle **MTN**

(7 marks)

b) Make y the subject of formula

$$y^{½} = \frac{W + W(xy)x^{½}}{x^{½}}$$

(8 marks)

END OF QUESTION PAPER